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Quantum Generative Models of Mid-Price Movement in Limit Order Books

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Abstract

In this article, we propose a quantum channel as a generative model of directional mid-price movements in a Limit Order Book (LOB). The observable stochastic evolution of the LOB's states is modeled as a time series of measurements of an 'ancilla' or 'emission' quantum system. The emission system is entangled with a second, unobservable quantum system representing the hidden dynamics of the LOB. The model is parameterized in the space of quantum circuits with mid-circuit measurements. While the proposed state process is Markovian, the observation process is generally not, allowing the model to capture both long- and short-term dependencies up to a fixed sequence length. The sequences of mid-price movement observations are formalized as a stochastic process language, defined by a set of finite distributions. A learning model and algorithm for these languages in the space of quantum channels are proposed. The quantum generative model is parsimonious, efficiently learnable, and analytically tractable. It provides analytical expressions of the probabilities of observed and future sequences of events. The implementation and the accuracy of the model are evaluated using historical LOB data. Our results demonstrate the potential of the quantum information processing models as a framework for studying of high frequency financial processes.

CCS Concepts

• **Computer systems organization** → *Quantum computing*; • **Theory of computation** → **Quantum computation theory**; *Evolutionary algorithms*; • **Mathematics of computing** → **Time series analysis**; **Stochastic processes**.

Keywords

Limit Order Books, QHMM, Quantum Generative Models

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1 Introduction

The markets of many financial assets are *order-driven markets*, where the orders to trade at particular price - the limit orders - are centralized in a *limit order book* (LOB). The *market orders*, requesting to trade immediately are matched and executed at the best available prices in the order book. The frequency of orders submission and the speed of execution are on the milliseconds scale. The order book is observable by all market participants in real time, serving as a source of high-frequency time series data. Models of these time series are critical for the success of trading and market-making algorithms [2, 4]. The models also contribute to fundamental theories of price formation and supply-demand dynamics [5, 25]. Empirical studies of the processes in order books at different observational frequencies reveal statistical properties such as transitional dynamics, roughness and persistence of volatility, correlations between prices, volumes, and volatility, long-memory dependencies, to mention a few [10, 16, 26]. This variety of "stylized facts" [9] leads to multiple modeling approaches.

The models of LOB dynamics can be divided into three categories: *agent-based models*, *statistical models*, and data-driven *deep learning* methods.

Agent-based models [7] describe market properties as a result of collective behavior among heterogeneous market participants such as market makers, institutional investors, high-frequency traders, and retail traders. In these models, agents act to maximize specific utility functions, and macro-level market dynamics emerge from the microstructure of their individual interactions.

Stochastic models describe the dynamics of the LOB either as a queuing system [5, 11] or through partial stochastic differential equations [13]. In the queuing system approach the state of the book is presented by sets of limit orders at specific price levels. The dynamics are defined by the impact of event queues, including order arrivals and cancellations. Order arrivals and cancellations are modeled as Poisson [11] or Hawkes [17] point processes. It is typically assumed that the LOB state evolves as a Markovian process[14].

Another group of stochastic models considers the evolution of order distributions in a continuous time framework. These models typically describe the order flow at all price levels [18]. The state of the book is defined by the volume of orders per unit price. The time evolution of the state is modeled as a spatial point process described by a stochastic partial differential equation.

There are multiple LOB models based on deep learning techniques [6, 27]. An end-to-end autoregressive generative model of the LOB is proposed in [22]. In [19], a generative model based on recurrent neural networks for the complete dynamics of a limit order book is developed. The model captures the LOB dynamics by

decomposing the probability of each transition into a product of conditional probabilities for order type, price level, order size, and time delay.

Several project study convolutional neural network architectures for modeling of the high frequency dynamics of a LOB. A convolutional approach for predicting of mid-price directional movements is proposed in [29]. An architecture with ReLU units that predicts mid-price trend and volatility, conditional on the state of the LOB, is studied in [15]. In [31], the authors present a deep learning architecture that combines convolutional layers to model dependencies within a LOB state and LSTM components to capture long-term dependencies between states. For a similar task in [30] the authors study a learning architecture with attention mechanism integrating Multilayer Perceptrons and LSTM.

In this article, we propose a quantum information view of the dynamics of the LOB, employing models of open quantum systems and quantum channels. In this approach, a non-observable quantum state evolves according to quantum dynamics and is entangled with an emission quantum system, which is measured at discrete moments. The measurement outcomes of the emission system provide classical information about the quantum state and simultaneously impact it due to their entanglement.

Similarly, market participants observe the state of the LOB and receive partial information about the hidden market state. Their behaviors depend on the observed information and, in turn, impact the market state.

The contribution of the articles are:

(1) A stochastic process language model of the LOB's mid-price movement.
(2) A quantum generative model of the LOB's mid-price movement.
(3) Analytically tractable expressions for the probabilities of observed and future movement sequences.
(4) A statistical approach for estimating the model's dimension from data
(5) Experimental results showing that the model effectively approximates the stochastic dynamics of a real LOB.

The organization of the article is as follows:

- **Section 2** defines three critical concepts discussed in this article: the mid-price and its directional movement process in an LOB, the directional process modeled as a stochastic process language, and quantum channels as generative models.
- **Section 3** presents the main steps of the learning process: empirical analysis and specification of the target process, estimation of model's complexity, the hypothesis space, the fitness function, and the learning algorithm.
- **Section 4** discusses the application of the method to real market data: data preparation, model order estimation, model training results.
- **Section 5** describes the application of the method on quantum computing device.
- **Section 6** provides conclusions and discusses current research directions.

2 Preliminaries

2.1 Limit Order Books

We consider an order-driven market for a financial asset. In such a market, participants may submit three types of orders: limit orders, market orders, and cancellation orders. The limit orders awaiting execution are collected in the limit order book. A limit order submitted at time t can be considered as a pair (p_t, v_t) where p_t is the specified price and $|v_t|$ are the units of the traded asset. If $v_t < 0$ the order is commitment to sell or a *sell order*, otherwise it is a *buy order*. The set of active orders in the order book at any moment of time t defines the full *state* of the book. The *bid price* p_t^b at time t is the highest price among the active buy orders at moment t and the *ask price* p_t^a is the lowest price among the active sell orders. The average of the bid and ask prices is called the *mid-price* p_t^m. Most trading activity occurs directly at the bid and ask prices, making the statistics of the mid-price critical for analyzing the LOB's dynamics [11, 12]. The *return* r_t of the mid-price is defined as

$$r_t = p_t^m - p_{t-1}^m \tag{1}$$

In this study we assume a model of the LOB, where the only observable is the *directional movement* of the mid-price defined as

$$y_t = \begin{cases} 1 & \text{if } r_t > 0 \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

We model the mid-price's directional movements using two time scales. In the first approach the times t are defined by events, such as order arrivals or cancellations. These event times are physically irregular. In the second approach, we consider physical times t at equal intervals of one second or longer. The mid-price within an interval is aggregated from the mid-prices during that interval. This aggregation can be done using various methods, such as taking the average, the weighted average, or simply the last mid-price within the interval.

2.2 Stochastic Model of Mid-Price Movement

We study the discrete-time stochastic processes of observed directional mid-price movements (2):

$$\mathcal{Y} = \{y_t : t \in \mathbb{N}, y_t \in \mathrm{A}\}, \tag{3}$$

where $\mathrm{A} = \{0, 1\}$ is referred to as an *alphabet*. The set of all finite binary sequences, including the empty sequence ϵ, is denoted by A^*. Finite sequences are indicated by the bold lowercase letters $\mathbf{a}$, $\mathbf{p}$, $\mathbf{s}$. The set of all sequences with length exactly t is designated by A^t. If $\mathbf{p}$ and $\mathbf{s}$ are sequences, then $\mathbf{ps}$ denotes their *concatenation*. The sequence $\mathbf{p}$ is called the *prefix* and $\mathbf{s}$ is called the *suffix* of $\mathbf{ps}$. For any $\mathbf{a} \in \mathrm{A}^*$ let $|\mathbf{a}|$ be the number of symbols in the sequence.

Any subset of A^* is a *language* L over the alphabet. The set of sequences originated by observations of the evolution of a mid-price directional movement process is called a *process language*. It is easy to verify that if a sequence results from the observation of a process, then every one of its subsequences has also been observed.

A *stochastic* process language L is a process language together with a set

$$D^L = \{D_k^L : k \geq 0\} \tag{4}$$

of finite dimensional probability distributions D_k^L, each of which is defined on the sequences with length exactly k:

$$D_k^L = \left\{ P[\mathbf{a}] : \mathbf{a} \in \mathrm{A}^k, \sum_{\mathbf{a} \in \mathrm{A}^k} P[\mathbf{a}] = 1 \right\} \tag{5}$$

Let's assume that the processes (3) are stationary. With any language L we can associate a *sequence function* defining the probabilities of realized sequences:

$$f^L : \mathrm{A}^* \to (0, 1) \tag{6}$$

defined as follows:

$$f^L(\mathbf{a}) = P[(y_{t+1}, y_{t+2}, \ldots, y_{t+k}) = (a_1, a_2, \ldots, a_k)], \tag{7}$$

where $\mathbf{a} \in \mathrm{A}^k, \mathbf{a} = a_1 a_2 \ldots a_k$

The sequence function f^L is defined by the distributions D_k^L (5) as follows:

$$f^{\mathbf{L}}(\mathbf{a}) = P[\mathbf{a}|D_k^L], \forall \mathbf{a} \in \mathrm{A}^k, \forall k \in \mathbb{N}. \tag{8}$$

The sequence function defines the probabilities of observations of all lengths (7). These probabilities capture the essential statistical properties of the process as distributions, joint distributions, conditional distributions.

The sequence function f^L is a complete definition of the stochastic structure of the process language L.

In the learning scenario, a stochastic language L is presented through observations of finite samples of its realizations (3). To estimate the state-space dimension of the language we consider a *bi-infinite* matrix, commonly referred to as a generalized *Hankel matrix*, defined by the sequence function (7) of the language [3]:

$$H^{\mathbf{L}} \in R^{\mathrm{A}^* \times \mathrm{A}^*}, H^{\mathbf{L}}[\mathbf{p}, \mathbf{s}] = f^{\mathbf{L}}(\mathbf{ps}), \forall \mathbf{p}, \mathbf{s} \in \mathrm{A}^*, \tag{9}$$

The matrix $H^{\mathbf{L}}$ is indexed by the prefixes $\mathbf{p}$ and suffixes $\mathbf{s}$ of the sequences $\mathbf{ps} \in \mathrm{A}^*$. The rows of $H^{\mathbf{L}}$ are indexed by the prefixes $\mathbf{p}$ in the *first lexicographical order*, and the columns are indexed by suffixes $\mathbf{s}$ in the *last lexicographical order* of the sequences [3]. The *rank* of a stochastic language is the rank of its Hankel matrix. A finite rank of the Hankel matrix is a necessary condition for the stochastic language to have a finite-state model as a HMM.

2.3 Quantum Generative Models

We propose a quantum generative model that defines the underlying stochastic dynamics of a process language as defined by the sequence function (8). The model is a *quantum operation* [24] or *quantum channel*, describing how a bipartite quantum system evolves according to its internal dynamics while part of it is observed by measurement. The model combines a unitary hidden state evolution with the emission of observations correlated with the hidden state. The model is defined as follows:

Definition 2.1 (Quantum Hidden Markov Model [21]). A *quantum* HMM (QHMM) $\mathbf{Q}$ over an N-dimensional Hilbert space $\mathscr{H}$ is a 4-tuple:

$$\mathbf{Q} = \{\mathrm{A}, \mathscr{H}, \mathcal{T} = \{T_a\}_{a \in \mathrm{A}}, \rho_0\}, \tag{10}$$

where

- A is a finite alphabet of observable symbols.
- $\mathscr{H}$ is an N-dimensional Hilbert space.
- $\mathcal{T}$ is a CPTP map (quantum channel) $\mathcal{T} : D(\mathscr{H}) \to D(\mathscr{H})$.
- $\{T_a\}_{a \in \mathrm{A}}, \sum_{a \in \mathrm{A}} T_a^\dagger T_a = I_N$ is an operator-sum representation of $\mathcal{T}$ in terms of a complete set of Kraus operators.
- ρ_0 is an initial state, $\rho_0 \in D(\mathscr{H})$.
- $D(\mathscr{H})$ is the space of density operators defined in $\mathscr{H}$.

The essential part of this definition is the *generative* character of the operator-sum representation of $\mathcal{T}$ which defines the system's dynamics in terms of Kraus operators corresponding to the observable symbols. When the operation $\mathcal{T}$ is applied to a state $\rho \in D(\mathscr{H})$, the model $\mathbf{Q}$ defines observation outcome or emission $a \in \mathrm{A}$ with probability:

$$P\big[a|\rho\big] = \mathrm{tr}(T_a \rho T_a^\dagger) \tag{11}$$

and the system's state after the emission becomes:

$$\rho_a = \frac{T_a \rho T_a^\dagger}{P\big[a|\rho\big]}. \tag{12}$$

The measurement probabilities in each state define a distribution:

$$D_\rho^{\mathbf{Q}} = \{P\big[a|\rho\big] : a \in \mathrm{A}\} \tag{13}$$

The model provides analytical expressions for the probabilities of the observed sequences. If the operation $\mathcal{T}$ is applied k times starting at the initial state ρ_0 a sequence $\mathbf{a} = a_1 \ldots a_k$ will be generated with probability

$$P\big[\mathbf{a}|\rho_0\big] = \mathrm{tr}(T_{\mathbf{a}} \rho_0 T_{\mathbf{a}}^\dagger). \tag{14}$$

and the final state will become

$$\rho_{\mathbf{a}} = \frac{T_{\mathbf{a}} \rho_0 T_{\mathbf{a}}^\dagger}{P\big[\mathbf{a}|\rho_0\big]}. \tag{15}$$

where

$$T_{\mathbf{a}} = T_{a_k} \ldots T_{a_1}$$

For each model $\mathbf{Q}$ the equation (14) defines a *sequence function* $f^{\mathbf{Q}}$ (6) as follows:

$$f^{\mathbf{Q}}(\mathbf{a}) = P\big[\mathbf{a}|\rho_0\big], \forall \mathbf{a} \in \mathrm{A}^*, \tag{16}$$

The sequence function (16) defines distributions $D_k^{\mathbf{Q}}$ over the sequences of length k:

$$D_t^{\mathbf{Q}} = \left\{ f^{\mathbf{Q}}(\mathbf{a}) : \mathbf{a} \in \mathrm{A}^t \right\} \tag{17}$$

Therefore every QHMM $\mathbf{Q}$ defines a *stochastic process language* $L^{\mathbf{Q}}$ (4) over the set of finite sequences A^*.

The eigenvector ρ^* of the operation $\mathcal{T}$ with eigenvalue of 1, normalized to represent a distribution, is the *steady state* of $\mathbf{Q}$:

$$\rho^* = \mathcal{T} \rho^*. \tag{18}$$

The model will define sequences of a stationary process when the initial state is the steady-state:

$$\rho_0 = \rho^*. \tag{19}$$

The dimension of the Hilbert space $dim(\mathscr{H}) = N$ is called *order* of the quantum model $\mathbf{Q}$.

The quantum HMMs offer more parsimonious representations of process languages than classical models. Specifically, it has been proven that if a stochastic process language can be represented by a hidden Markov model (HMM) with n states, there exists a quantum hidden Markov model (QHMM) for the same language with $\sqrt{n}$ states [21].

3 Learning Quantum Generative Models of Directional Mid-Price Movement in an LOB

This section formalizes the learning process for quantum generative models of mid-price directional movement in LOBs.

3.1 Empirical Specification of Mid-Price Movement Process Language

In the learning scenario, the studied mid-price movement stochastic language is presented through one or more finite samples of the process (3). These samples are observed directly on a particular electronic trading platform. The sequence distributions (4) defining the unknown process language need to be estimated from these observations. If we assume, that the real data generation process has finite order n, then its finite dimensional distributions D^L are completely defined by the values of the sequence function $f^L(\mathbf{a})$ for all sequences $\mathbf{a}$ of length at most $2n-1$ [8]. The subset of the Hankel matrix (9) with dimension $n \times n$ would have rank n. Hence, the exact specification of the model of rank n requires a sample of sequences and respective probabilities with size $m = \sum_{i=1}^{2n-1} 2^i$. The estimate uses the rank of the unknown language, but in most learning scenarios, the rank is not provided, and it is not even known whether such a finite rank exists. Instead, the available training data consists of a set of observed sequences of length T sampled from the unknown process language L (3). Using the empirical samples we have to estimate the rank and derive an approximation $\hat{f}^L$ of the sequence function (8). For sequences with length k we construct the set:

$$\mathbf{S}_k = \left\{\mathbf{a} : |\mathbf{a}| = k, \exists \mathbf{p}, \mathbf{s} \in \mathrm{A}^*, \exists y \in \mathcal{Y}_L : \mathbf{pas} = \mathbf{y}\right\}. \tag{20}$$

If the sample $\mathbf{S}_k$ contains l independently drawn sequences from the unknown distribution D_k^L, then the estimate $\hat{f}^L(\mathbf{a})$ is the empirical frequency of $\mathbf{a}$ in $\mathbf{S}_k$:

$$\hat{f}^L(\mathbf{a}) = \frac{1}{l} \sum_{\mathbf{b} \in \mathbf{S}_k} \mathbf{1}_{\mathbf{a}}(\mathbf{b}) \tag{21}$$

and the estimate of the finite distribution D_k^L is

$$\hat{D}_k^L = \left\{\hat{f}^L(\mathbf{a}) : \mathbf{a} \in \mathbf{S}_k\right\} \tag{22}$$

If we are given a sample $\mathcal{Y}^L$ with no additional information we have to estimate a plausible order of the model by determining the maximum rank of the Hankel matrix. An intuitive procedure involves starting with short sequence lengths and incrementally increasing the length while estimating the rank of H^L. If the rank stabilizes and does not increase further at a particular sequence length, we can use this rank as an estimate. However, this procedure will only provide a local approximation, as inferring the rank of the Hankel matrix from data is an undecidable problem [28].

3.2 Finite Approximation of the Sequence Function

In practical situations, we may assume that the Hankel matrix has a rank n, but consider this rank to be too high for various reasons: the state space may be computationally intractable, the probabilities in the matrix may be very low and not correspond to plausible events, or many of the singular values may have diminishing amplitudes. In these cases, we can use a low-rank approximation H_m^L of H^L accounting only for the largest m of its singular values, and learn the model in a space with dimension $m < n$. Let's assume that the singular values are ordered as $\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_n > 0$. Since the sum of the squared singular values represents the total variance (information) in a matrix H^L, the variance approximation error for its low-rank approximation H_m^L will be

$$\epsilon_m = \frac{\sum_{i=m+1}^{n} \sigma_i^2}{\sum_{i=1}^{n} \sigma_i^2}. \tag{23}$$

The total probability approximation error is restricted by the rank-m variance approximation error:

$$\sum_{\mathbf{p},\mathbf{s} \in \mathrm{A}^*} \left| H^L[\mathbf{p},\mathbf{s}] - H_m^L[\mathbf{p},\mathbf{s}] \right|^2 \leq \epsilon_m \sum_{\mathbf{p},\mathbf{s} \in \mathrm{A}^*} \left| H^L[\mathbf{p},\mathbf{s}] \right|^2 \tag{24}$$

If the rank of a stochastic process language is infinite, but the sum of the squared singular values of its Hankel matrix converges, then the matrix has finite Frobenius norm . This norm can be estimated, and the finite model approximation error can be controlled by selecting an appropriate number of singular values m. If the stochastic process language has infinite rank and the sum of the singular values does no converge, it generally means that the process cannot be adequately modeled by a finite-dimensional model. In such cases, the finite-rank approximation of the Hankel matrix can still provide an approximation of probabilities for sequences of restricted length.

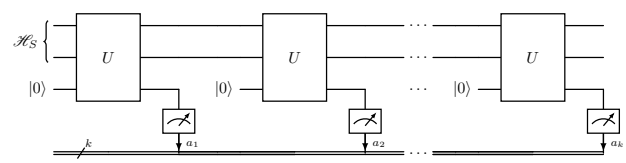

Figure 1: Implementation of QHMM

3.3 Hypothesis Space of Unitary Quantum Generative Models

The definition of a quantum stochastic generator in the previous section is based on the concept of a quantum channel acting on a quantum state. This framework provides a theoretical base to approach quantum stochastic generators as channels for simultaneous quantum and classical information processing. However, this approach cannot be directly implemented on quantum computing hardware. Therefore, we consider parameterizing and learning of QHMMs within the space of quantum circuits with mid-circuit measurement.

Definition 3.1 (Unitary Quantum HMM [21]). A *Unitary Quantum* HMM $\mathbf{Q}$ over a finite alphabet of observable symbols A and finite N-dimensional Hilbert space is a 6-tuple:

$$\mathbf{Q} = \left\{\mathrm{A}, \mathscr{H}_S, \mathscr{H}_E, U, \mathcal{M}, R_0\right\}$$

where

- A is a finite set of m observable symbols.
- $\mathscr{H}_S$ is the Hilbert space of the hidden states of dimension N.

- $\mathscr{H}_E$ is the Hilbert space of an auxiliary emission system with dimension $m \leq M \leq N^2$ and basis $E = \{|e_i\rangle\}_{i=0}^{M-1}$.
- U is a unitary operator defined on $\mathscr{H}_S \otimes \mathscr{H}_E$.
- $\mathcal{M}$ is a bijective map $\mathcal{P}_m^E \to \mathrm{A}$, where $\mathcal{P}_m^E$ is an m-element partition of E.
- $R_0 = \rho_0 \otimes |e_0\rangle \langle e_0|$ is an initial state.

A Unitary QHMM is fully defined by the operator U, map $\mathcal{M}$ and initial state R_0. An implementation of UQHMM where $\rho_0 = |s_0\rangle \langle s_0|$ is presented in Figure 1.

The unitary physical implementation of a quantum channel defines a generative model (16) and a stochastic process language $L^{\mathbf{Q}}$ (17) over the set of finite sequences A^* since the the quantum channel $\mathcal{T}$ is fully specified in the terms of the unitary transformation U as follows:

$$\mathcal{T}\cdot = \sum_{a\in \mathrm{A}} T_a \cdot T_a^{\dagger}. \tag{25}$$

where

$$T_a = \prod_{e\in E_a} K_e, \tag{26}$$

$$\mathcal{M}(E_a) = a, \tag{27}$$

$$K_e = \left(I^N \otimes \langle e|\right) U \left(I^N \otimes |e_0\rangle\right). \tag{28}$$

Given a set of samples $\mathcal{Y}^L$ (3) from a directional mid-price movement process L the hypothesis space of the quantum generative models of L is a set of unitary QHMMs (3.1) defined as a triple:

$$\{\mathbf{U}, \mathbf{S}, \mathrm{M}\}, \tag{29}$$

where the set $\mathbf{U}$ contains state-transition unitary operators on the Hilbert space $\mathscr{H}_S \otimes \mathscr{H}_E$, the set $\mathbf{S}$ contains steady-state preparing unitary operators on the Hilbert space $\mathscr{H}_S$, and M is the set of bijective maps $\mathrm{A} \to \mathrm{A}$. The steady-state preparation operators transform a fixed maximally mixed state ρ_0 to a specific for each learning target L steady-state ρ^*. The dimension N of the Hilbert space $\mathscr{H}_S$ is estimated from the data $\mathcal{Y}^L$, and for binary observations the dimension of $\mathscr{H}_E$ is 2. Every state-transition or state-preparation unitary operator has representation as a quantum circuit C_U :

$$C_U = (g_i)_{i\geq 1}. \tag{30}$$

The quantum gates g_i are encoded as 3-tuples:

$$g = \langle \mathbf{t}, ([q_c,]q_d), ([p_1[, p_2]])\rangle, \tag{31}$$

where $\mathbf{t} \in \mathcal{G}$ is the gate's type, q_c and q_d are the control and data qubits, and p_1, p_2 are gate's parameters. We will make the assumption that the set of gate types $\mathcal{G}$ includes single-qubit gates and two-qubit controlled gates [24]. Formally, the hypothesis space is the infinite set of all finite quantum circuits in the $2\times N$-dimensional Hilbert space $\mathscr{H}_S \otimes \mathscr{H}_E$, encoded as lists of gates from $\mathcal{G}$ (31). The hypothesis space is parameterized by the circuits sets $\mathbf{U}$ and $\mathbf{S}$:

$$\mathcal{Q} = \left\{Q(U,S) : U \in \mathbf{U}, S \in \mathbf{S}\right\}. \tag{32}$$

The generative model corresponding to a hypothesis $Q(U,S)$ is specified by the string function (14),(16):

$$f^{Q(U,S)}(\mathbf{a}) = P\left[\mathbf{a}|\rho^*\right], \forall \mathbf{a} \in \mathrm{A}^*, \tag{33}$$

where $\rho^* = S\rho_0 S^{\dagger}$.

In the rest of the article, to simplify the notation, we assume that the steady-state preparing unitary S is constant, and we refer to the hypothesis defined by the state-evolving unitary U by $Q(U)$ or simply Q when there is no ambiguity.

3.4 Quality of a Hypothesis

A standard measure of the quality of a generative hypothesis Q is its variational distance to the target distributions D^L(4). The probabilistic divergence between the target and the hypothesis for any sequence $\mathbf{a} \in \mathrm{A}^*$ can be defined as:

$$\delta^{LQ}(\mathbf{a}) = \left|\hat{f}^L(\mathbf{a}) - f^Q(\mathbf{a})\right|. \tag{34}$$

The variational distance between the target language and the hypothesis on finite sequence distributions with lengths n is

$$\Delta_n(D^L||D^Q) = \max_{\mathbf{a}\in \mathrm{A}^n} \delta^{LQ}(\mathbf{a}) \tag{35}$$

The average variational distance between the hypothesis and the target given sample of sequences with lengths up to n is defined as

$$\Delta_{\leq n}(D^L||D^Q) = \frac{1}{n}\sum_{i=1}^{n} \Delta_i\left(D^L||D^Q\right) \tag{36}$$

We will refer to the average distance (36) as the *empirical divergence* between a hypothesis and the learning target.

The empirical divergence (36) is used as a hypothesis quality function called a *fitness function*:

$$F(Q,n) = -\Delta_{\leq n}\left(D^L||D^Q\right), \tag{37}$$

For the sake of simplicity, going forward, we assume that the size of the sample n in $F(Q,n)$ is a constant, and we will use $F(Q)$ instead.

3.5 Continuity of the Fitness Function

It is important to note, that the fitness function (37) has properties which imply efficiency of the learning process. Let D^1 and D^2 be sequence distributions defined by the hypotheses $\mathbf{Q}(U_1)$ and $\mathbf{Q}(U_2)$. In [21] it is shown that the average empirical divergence (36) of these distributions is restricted by the operator norm of the difference between the corresponding unitaries:

$$\frac{1}{2n}\Delta_{\leq n}\left(D^1||D^2\right) \leq \Delta U^{12}, \tag{38}$$

where

$$\Delta U^{12} = \|U_1 - U_2\|. \tag{39}$$

This result suggests that the difference of the fitness values of two hypotheses $Q^1 = \mathbf{Q}(U_1)$ and $Q^2 = \mathbf{Q}(U_2)$ is dominated by the operator norm of the difference of the corresponding unitaries:

$$\frac{1}{2n}\left|F(Q^1) - F(Q^2)\right| \leq \Delta U^{12}, \tag{40}$$

This result indicates continuity of the fitness function with respect to parameters in the hypothesis space, a property that is crucial for designing efficient learning algorithms.

3.6 Quantum Generative Model Learning Algorithm

An algorithm for learning QHMMs is proposed in [21]. Here, we discuss its application to learning the introduced models of directional mid-price movements in Limit Order Books (LOBs). The input to the algorithm includes a training dataset, a unitary hypothesis space,

and a fitness function. The goal of the algorithm is to find a hypothesis—a generative model—that maximizes the fitness function (37). Every hypothesis $Q(U) = (S_U, C_U)$ comprises both discrete (structural) and continuous components. The discrete component S_U is a sequence of gate types and the qubits they are related to as defined in (30, 31). The structural components define a combinatorial search space. The continuous components are defined by vectors P_U containing gates' parameters:

$$P_U = [p_i : p_i = g_i[2], C_U = (g_i)_{i \leq 0}],$$

where $g_i[2]$ is the parameters vector of i-th gate of C_U (31). For optimization in this hybrid search space we use a *memetic algorithm* [23] that combines evolutionary global search and *Metropolis annealing* local search in the structural subspace with non-linear parametric optimization in the parameters subspace. The fitness of every structure S_U is estimated by parametric optimization(37):

$$P_U^* = \underset{P_U}{\text{argmax}} F(Q(S_U, P_U)),$$

performed by a set of non-linear solvers. The type of solver used by any hypothesis is encoded in its representation is subject to self-adaptation through the main evolutionary process.

The algorithm starts with a finite sample of μ random hypotheses drawn uniformly from the hypothesis space $\{\mathbf{Q}\}$. These hypotheses form the initial *parent population*. Subsequently, the algorithm implements an iterative population evolving process comprising three randomized steps: selection of a set of parents for offspring generation, offspring generation by applying variation operators to the parents, and selection of parent and offspring hypotheses to survive in the next generation. (**Algorithm** 1).

Algorithm 1: QHMM Evolutionary Learning

Input:
- μ - size of parent population
- λ - size of offspring population
- $\{D_t^L : 0 < t \leq n\}$ - sample of a learning target
- $\hat{N}, M$: dimensions of hidden states and emission systems
- $\mathcal{Q}$ (29) - hypothesis space
- $F(Q)$ (37) - fitness function
- $\mathcal{G}$ - quantum gates set
- t_{max} - maximal number of evolution generations

Output: Best hypothesis found: $Q^* = \underset{Q \in \mathcal{Q}}{\text{argmax}}\ F(Q)$

```
t ← 0                                                   ▷ Generations counter
P_t ← {P_t[i] ~ Uniform(Q) : i ∈ [1, μ]}                ▷ Initial population
S_t ← []                                                ▷ Search state variables
D_t ← Uniform(.)                                        ▷ Search distributions
while t < t_max do
    τ ← Temperature(t)                                  ▷ Local Metropolis search control
    γ ~ D_t(γ)                                          ▷ Adaptive offspring-parents ratio
    λ_t = γ · μ                                         ▷ Offspring size
    O ← {O[i] ~ D_t(Selection, P_t) : i ∈ [1, λ]}       ▷ Selection of parents
    O ← {O[i] : O[i] = Modify(O[i], τ), i ∈ [1, λ]}     ▷ Offspring generation
    P_{t+1} ← {P_{t+1}[i] : P_{t+1}[i] ~ D_t(Survival, P_t ∪ O)}  ▷ Survivors selection
    S_t ← State(P_t, S_{t−1}, ...)                      ▷ Search state update
    D_{t+1} ← RL(D_t, S_t)  ▷ Reinforcement Learning of Search Control Distributions
    t ← t + 1
return argmax_{t ≤ t_max, i ∈ [1,μ]} F(P_t[i])
```

A complete specification of the evolutionary search algorithm used for learning quantum generative models is provided in [21].

4 Examples of Learning Models of Mid-Price Movement in LOBs

We created generative models of directional mid-price movement at event level data and data integrated at one second interval.

4.1 Data Preparation

Our datasets consist of data at the LOB event level for Amazon.com Inc. (AMZN) on 06/03/2022 and Alphabet Inc. (GOOGL) on 07/19/2022. The size of each of the time series is $T = 102418$ events for AMZN and $T = 235928$ events for GOOGL. These daily data sets define sequences of directional movement of the mid-price $\{p_k^m\}_{k=0}^T$ at event time $\{t_k\}_{k=0}^T$. The learning samples for generative models at event time are created by calculation of the mid-price returns (1) and directional movements (2).

We also create models of the mid-price movements aggregated over physical time interval of one second. The mid-price for each interval is represented by the mid-price of the last event in the interval. The returns and the directional movements are calculated in the same manner as in the event-time case. The learning algorithm requires calculation of the fitness function (37), which depends on the finite distributions (22). The distributions are estimated by the empirical samples (20), constructed from the daily time series of the directional movements.

4.2 Model Order Estimation

To estimate the order of the process language and the dimension of the Hilbert space of the model for both event-time and aggregated observations, we follow the procedure described in Section 3.1. The rank of the finite Hankel matrices were estimated for sequences with length op to 24 symbols. Longer sequences appear with low frequencies and their probabilities cannot be estimated. The estimations presented in Figure 2 show that a 3-qubit quantum system simulating a stochastic language of order 64 can accurately define the probabilities of sequences with lengths up to 10 for the 1-second model and up to 12 for the event-time model (indicated by the yellow dashed horizontal line). If we increase the dimension of the quantum system by 2 qubits, the model can accurately define the probabilities of sequences with lengths up to 20 for the event-time case and up to 18 for the process aggregated at 1-second intervals.

It is evident from the graph, that the rank of the Hankel matrices for both event-time and aggregated movements at 1-second intervals grows exponentially with the size of the sequences. This behavior suggests a non-finite state stochastic process, which is common for both modeled assets. If we allow a total approximation error of 1%,(23) or $\epsilon = 0.01$ (24), we can model significantly longer sequences using small quantum models, as shown in Figure 3. A quantum model of 3 qubits can approximate the probabilities of sequences with lengths up to 14 for the 1-second aggregated process and up to 16 for the event-level process. For a 5-qubit model, these lengths are 20 and 22, respectively.

4.3 Model Training

We train the models in the hypothesis space (29) with parameters defined by (32). All models are created in quantum Hilbert spaces

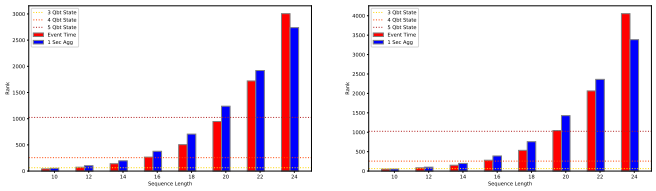

Figure 2: Rank of Hankel matrices by sequence length. Left: GOOGL. Right: AMZN. The horizontal lines mark the ranks of 2,4, and 5 qubit quantum models

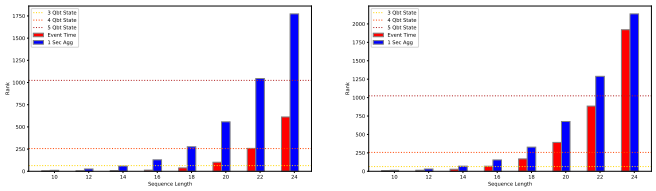

Figure 3: Low-rank approximation of Hankel matrices by sequence length. Precision of the approximation $\epsilon = 0.01$. Left: GOOGL. Right: AMZN.

with dimension $N = 8$ (3 qubits). The mid-price movement processes are specified by distributions (22) of sequences with lengths up to 10 symbols. The quantum gate set (30) contains single qubit gates $\{p, x, y, z, h, rx, ry, rz\}$ and their controlled two-qubit counterparts [20]. As initial state of the model ρ_0 we use the maximally mixed state implemented by the initial segment of circuits shown on Figure (6). The map $\mathrm{M} : \{0, 1\} \longrightarrow \{0, 1\}$ (29) is implemented by the presence or absence of an x gate as last operation on the measurement qubit.

The search for a hypothesis that maximizes the fitness function is implemented using the evolutionary process presented in Algorithm 1. The algorithm uses a parent population of size $\mu = 200$ hypotheses. Each generation produces an offspring population of size $\lambda = \gamma\mu$, where γ is adapted hyper-parameter controlling exploration/exploitation trade-off of the search. The "Modify"variation operators and the the meta-level control of distributions and hyper-parameters, such as temperature, are implemented as described in [21]. The maximum number of evolutionary generations was set to 300 generations or until the fitness of the best individual exceeds -10^{-4}. For the specific number and lengths of sequences, this is the maximal precision of a sequence probability defined by the sample.

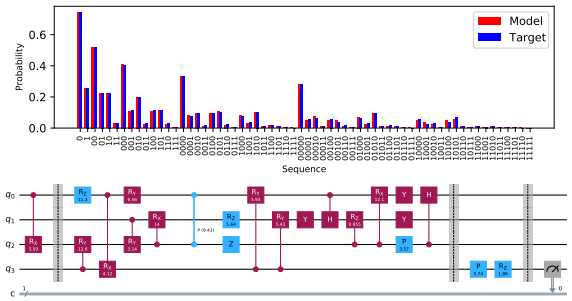

Figure 4: Model of Mid-Price Sequence Distributions in event time: GOOGL. Upper: Empirical and Model Distributions. Lower: Quantum Algorithm

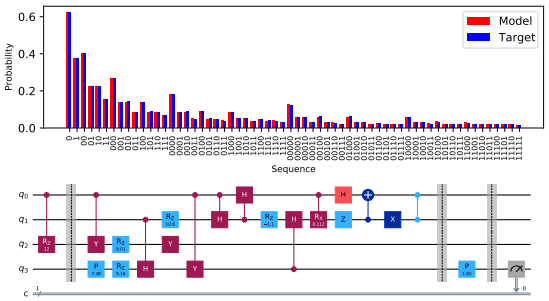

Figure 5: Model of Mid-Price Sequence Distributions in 1 second: AMZN. Upper: Empirical and Model Distributions. Lower: Quantum Algorithm

4.4 Results

For each data set described in **Section 4.1**, we learned models at event time and at aggregated 1-second time periods. In all four cases, the variational distance between the model and target distributions was optimized to values less than 10^{-4} (35). The target and model-defined sequence distributions for the GOOGL LOB in event time are presented on the left side in Figure 4. The figure on the right presents the quantum algorithm. The barriers separate the steady-state preparation, transitional dynamics, and measurement parts of the algorithm. The results for AMZN in 1-second aggregated intervals are presented in Figure 5.

5 Quantum Hardware Experiments

The circuits trained to model AMZN and GOOGL sequences as implemented in Qiskit [20] are shown on Fig. 6. Both circuits are build for 4 qubits, however, it is easy to notice that circuit (a) requires almost full connectivity and circuit (b) requires full connectivity between qubits. In order to run this on hardware one would need to transpile the circuit to use the set of native gates for the quantum processor (QPU) and comply with its layout. In order to control the depth of the circuits we will focus on 3 time steps of the processes. The circuits (a) and (b) for 3 time intervals transpiled to native gate sets without optimization have the 2-qubit depths of 100 and 110, respectively. By using multi-stage transpilation process and various optimization techniques we were able to reduce the depths to 53 and 86, respectively. The optimization of circuit (b) is less successful due to higher qubit connectivity requirement. We have run these

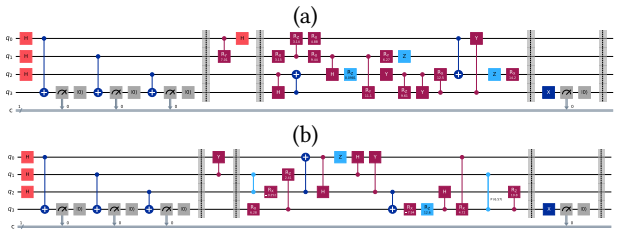

Figure 6: Trained circuits modeling AMZN (a) and GOOGL (b) for 1 time step. The first segment is initial state, the second segment - initial state preparation, third segment - evolution unitary, fourth segment - measurement. The third and fourth segments are repeated for each time step.

circuits on IBM Eagle r3 QPU using dynamical decoupling error

suppression and Pauli Twirling error mitigation techniques. Details on the transpiled circuits and hardware run results can be found here [1]. The final distributions for 3-symbol sequences are shown on Fig. 7.

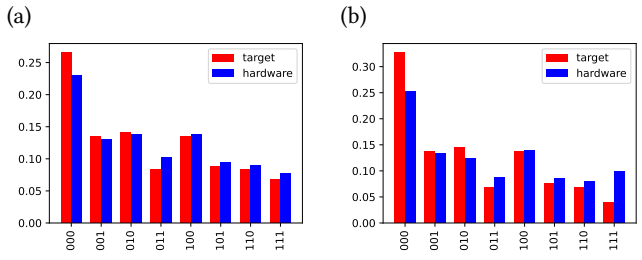

Figure 7: Hardware results for (a) AMZN and (b) GOOGL sequences.

Histograms demonstrate good agreement between the observed target distribution and the one modeled by the hardware, though some variance is evident due to noise. This variance is more prominent in Figure 7(b), as it corresponds to a deeper quantum circuit.

6 Conclusion and Outlook

In this article, we employed a class of quantum generative models - QHMMs [21], to reproduce the directional movements of the mid-price in LOBs. Quantum HMMs offer two significant advantages over their classical counterparts. Firstly, they exhibit quadratically better parsimony capabilities [21]. Secondly, they can capture short-term dependencies because the hidden state system and the emission system are quantum entangled. The outcome of a measurement depends on the current quantum state and also impacts the next state through entanglement. We demonstrated that the proposed generative models can be effectively trained to simulate the daily mid-price movements of the LOB, both at the event level and at aggregated intervals. Given the complexity and noise inherent in LOB dynamics, we also discussed methods for their low-rank approximation. We have optimized the final circuit for quantum hardware and tested different methods for error suppression and mitigation. The output from the quantum hardware has successfully simulated the observed distributions. The performance can be further improved by using hardware-efficient ansatz circuits. Current research directions include applying this approach to larger sets of data and evaluating the performance of the models on out-of-sample scenarios.

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